Oxford Development Studies

ISSN: 1360-0818 (Print) 1469-9966 (Online) Journal homepage: www.tandfonline.com/journals/cods20

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To cite this article: Noman Ahmad, Faiz Ur Rehman & Nasir Sarwar (2023) COVID-19 induced national lockdown and income inequality: evidence from Pakistan, Oxford Development Studies, 51:1, 66-81, DOI: 10.1080/13600818.2022.2091124

To link to this article: https://doi.org/10.1080/13600818.2022.2091124

Published online: 20 Jun 2022.

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OXFORD DEVELOPMENT STUDIES
2023, VOL. 51, NO. 1, 66–81
https://doi.org/10.1080/13600818.2022.2091124

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COVID-19 induced national lockdown and income inequality: evidence from Pakistan

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ABSTRACT
COVID-19 posits two risks to developing countries. On the one hand, growing COVID-19 cases exposed the vulnerabilities of the already debilitating health sector, while, on the other hand, policies to control the spread of COVID-19 can exasperate economic disparities. In this article, we examine one such policy response to control the spread of COVID-19 by the Government of Pakistan, the National Lockdown. This study assesses a plausible impact of this policy response on income equality across Pakistan. By exploiting a nationally representative household survey, it is observed that COVID-19 induced national lockdown is associated with increased income inequality in Pakistan. Our estimates show that about a 16 percent increase in income differences between the top 10 and the bottom 10 percent of the population could be associated with national lockdowns. Gini Coefficient also indicates an up to 3 percent increase in inequality after lockdown. At the disaggregated level, the suggestive evidence shows that inequality has increased within the urban population. Interestingly, the lockdown has also increased the inequality within occupations that can be managed remotely from home.

KEYWORDS
COVID-19; national lockdown; income inequality; Pakistan

Introduction

COVID-19 distressed the world in multiple ways because of its direct and indirect effects on people's lives. This pandemic, or any other, unswervingly affects the health sector by putting immense pressure on the health facilities by increasing the demand for already limited resources and health-care professionals. The gravity of COVID-19, however, is not limited to the health sector. Given the nature of the pandemic, the initial recommended action plan to control the spread of the virus included movement and interactive restrictions or lockdowns (World Health Organization, 2020a). Restriction on movement and interaction or adoption of physical distancing and lockdowns, generally hamper economic activity (Caraka et al., 2020; Hossain, 2021; Mandel & Veetil, 2020; Sahoo & Ashwani, 2020; Salik & Rafique, 2020; Shafi, Liu, & Ren, 2020). The World Bank (2020) estimates a shrinkage of 5.3% in global output in 2020. Moreover, the United Nations Department of Economic and Social Affairs (2020) predicts a sharp rise in poverty and inequality within and between countries.

The pandemic not only distresses the developing countries unevenly but the economic recovery in these countries is also sluggish (UNCTAD, 2020). The already inadequate capacity, limited resources, the nature of jobs, lack of digitalization, a large share of daily wage earners in the labor force, and a sharp increase in unemployment could have contributed to the socio-economic

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adversities of the pandemic-induced lockdowns in developing countries. Lockdowns and fraught supply chains disrupted business activity in Pakistan, while a lack of labor contracts and legal protection exacerbated layoffs in the labor market (Human Rights Watch, 2020). Accentuating that the pandemic-induced lockdown has a disproportionate effect on marginalized segments of the labor force in Pakistan (Human Rights Watch, 2020; UNDP, 2020), this paper examines its impact on inequality and contributes to the growing literature addressing the impact of COVID-19.

This paper utilizes a nationally representative survey, 'Special Survey for Evaluating Socio-Economic Impact of COVID-19 on Wellbeing of People,' conducted by the Pakistan Bureau of Statistics (2020). The survey collects comprehensive information about employment and income prospects, among other important socioeconomic indicators, from households before, during, and after COVID-19 induced lockdowns. Utilizing this data, percentile dispersion ratios, Gini coefficient, and Atkinson's inequality measure have been estimated to assess income inequality in Pakistan. This study also provides inequality statistics at the disaggregated level. Furthermore, the survey data is matched with the Pakistan Standard Classification of Occupations (PSCO) to pronounce the effect of COVID-19 on different occupational categories.

The findings of this work, in line with (Blundell, Costa Dias, Joyce, & Xu, 2020; Elgar, Stefaniak, & Wohl, 2020; Galletta & Giommoni, 2022; Hanspal, Weber, & Wohlfart, 2020; Qian & Fan, 2020; Sayed & Peng, 2021; Witteveen, 2020), suggest that COVID-19 induced shutdowns contribute to a more skewed income concentration (estimated by percentile dispersion ratios). Our findings suggest an increase in income inequality across Pakistan, with up to 16% increase in income differences between the top 10% and the bottom 10% of the population, could be associated with the national lockdowns. The value of the Gini coefficient also rose from 0.439 to 0.453 indicating an increase of 0.014 (3% increase in the level of inequality). Atkinson's Inequality measure using three different values of sensitivity parameter also suggests an increase in income inequality. The disaggregated analysis indicates that inequality during lockdown increased only in Punjab and Balochistan but remained the same in KPK and Sindh. Furthermore, inequality is observed to have been aggravated in urban areas given the high exposure of urban areas to the virus (Du, King, & Chanchani, 2020) and hence the disruption of business activities in these areas. Our findings contradict (Clark, d'Ambrosio, & Lepinteur, 2021) who show that COVID-19 reduced inequality in five European countries (France, Germany, Italy, Spain, and Sweden).

Background and timeline of lockdown restrictions

Pakistan officially reported its first COVID-19 case on 26 February 2020.[1] Since then, we witnessed an exponential increase in COVID-19 cases in Pakistan. The official website of the Government of Pakistan reported 534,041 confirmed cases as of 25 January 2021. Amongst the provinces, Sindh leads in the absolute number of confirmed cases (241,200), Punjab ranks second (154,017), and then comes KPK (65,532), Islamabad (40,815), Baluchistan (18,750), and Gilgit-Baltistan. Pakistan had started to prepare against the pandemic in January 2020 when WHO headquarters sent technical guidance to Pakistan. According to World Health Organization (2020b), Pakistan's fight against coronavirus has been a 'welcome relief.' Support from international agencies especially WHO got Pakistan to a strong start against fighting coronavirus.

To restrict the escalating coronavirus cases, a series of smart lockdowns were imposed which range from shutting down markets to closing borders and restricting international flights (Shahzad & Hassan, 2020). Following many other countries, Pakistan's federal and provincial governments imposed a nationwide lockdown from April 2020. The lockdowns may have helped to flatten the coronavirus cases curve but dented the already struggling economy.

The series of lockdowns in Pakistan started when the Government of Pakistan shut its international western borders and closed all the educational institutions in the country on 13 March 2020, the day when the country reported its first locally transmitted COVID-19 case. On 23 March 2020, Punjab and Sindh governments announced the first lockdown having the largest coronavirus surge among

provinces. The province of Baluchistan followed the two provinces and imposed a lockdown the next day. National Coordination Committee (NCC), with the unanimous support of all the provinces, extended the lockdown restrictions to 14 April 2020. The national command and Operation Center (NCOC) was founded on 20 April 2020 to monitor the national COVID-19 situation and to form and implement policies (from administering the vaccine to implementing lockdown restrictions). The sequence of lockdowns continued until the end of July, but there were exceptions during this period. Shopping malls and business activities were allowed to operate for a given time during the day, especially during the holy month of Ramadan. Keeping in view the political and economic implications of national lockdown, the government instead shifted its preferences towards micro-level lockdown (called smart lockdown), which focused on areas with higher positivity ratios rather than restricting all activity at the provincial or national levels (The Express Tribune, 2020).

During the pandemic, fiscal accounts have come under tremendous pressure, budget targets deviated, and the GDP growth rate contracted to a negative 0.4% (Pakistan Economic Survey, 2020). A large number of workers lost their jobs during the pandemic and according to Planning Commission, the informal sector took a heavy toll. The commission projects an increase in the number of unemployed and estimates the unemployment rate for the year 2020–2021 to be 9.56% (Planning Commission, Ministry of Planning, Development and Special Initiatives Government of Pakistan, 2020). Human Rights Watch (2020) highlighted that the pandemic and the responsive measures have disproportionate 'consequences for garment and textile workers, domestic workers, home-based workers, and other workers in low-income households.' According to UNDP (2020), people who earn daily wages and who live in the poorest wealth quantile are among the most vulnerable groups. All these statistics indicate how badly the pandemic has disturbed Pakistan's macroeconomic outlook, therefore, we extend this debate by estimating the effects of the pandemic at the microeconomic level.

Data and methodology

To evaluate the effects of COVID-19 on income inequality, this study exploits the latest nationally representative survey, 'Special Survey for Evaluating Socio-Economic Impact of COVID-19 on Wellbeing of People,' administered by the Pakistan Bureau of Statistics (2020). The survey is based on a Two-stage Stratified Random Sampling Design in which, at the first stage, 500 Principal Sampling Units (PSUs) are selected using Systematic Random Sampling. Due to the relevance of COVID-19 related lockdowns to urban regions, 70% of the sampled PSUs are urban while the remaining 30% are comprised of rural areas. At the second stage, 12 Households within each PSU are selected using the Systematic Random Sampling Technique. The survey collects comprehensive and representative information regarding employment, food security, coping strategies during the national lockdown, mitigation measures, and reduction in the use of healthcare due to COVID-19.

The study exploits the information related to employment and the average monthly income of adult respondents to estimate different indices of inequality. The question most relevant to our income inequality indices is the Average Monthly Income of respondents before National Lockdown due to the first wave of COVID-19, and after major lock-down restrictions were rescinded. Section B of the survey, 'Impact of COVID-19 on Employment and Income' asks the respondents about their average monthly income before (April 2020), during (April–July 2020), and after (July 2020) national and provincial COVID-19 lockdowns. It is worth mentioning that data are recalled and not panel, each respondent has been asked to report his/her income before, during, and after the lockdown at the time of response (in July–August 2020)[2].

This analysis also provides disaggregated results for income inequality at different levels. We also estimate the inequality across different occupations that could be done from home during the lockdown. To determine the effects of COVID-19 lockdown on specific industries, we match the survey data with the Pakistan Standard Classification of Occupations (PSCO). Following are the details of measures we used in our analysis to estimate income inequality.

While estimating the income inequality through this method, we compare the monthly income earned by different groups. For instance, we take a ratio of the richest 10% of the population to the poorest 10% of the population to estimate p90/p10 ratio. Similarly, we also estimate inequality by comparing the top 10% and the bottom 50% of the population (represented by p90/p50); the top 25% of the population, and the bottom 25% of the population (shown as p75/p25). Second, we use the Gini coefficient to estimate income inequality at different levels. Gini coefficient estimates inequality at the scale between 0 and 1 where a higher value means higher inequality among the population. Figures 1 and 2 depict the geographical distribution of inequality at the sub-national level measured using the Gini coefficient. Third, we use Atkinson's Inequality measure to gauge income inequality. In his book, *The Economics of Inequality*, Atkinson stated that an inequality 'cannot, in general, be measured without introducing social judgments. Measures such as the Gini coefficient are not purely "statistical" and they embody implicit judgments about the weight to be attached to inequality at different points on the income scale' (Atkinson, 1983, p. 47). Keeping this in consideration, Atkinson's Inequality index includes a sensitivity parameter (ε). This parameter ranges from 0 (indicating that the researcher is indifferent about the nature of the income distribution), to infinity (where the researcher is concerned only with the income position of the lowest income group) (De Maio, 2007, p. 850). Mathematically, the index can be represented as follows:

$$A_\varepsilon = 1 - \frac{1}{\mu}\left[\frac{1}{N}\sum_{i=1}^{N}(y_i)^{1-\varepsilon}\right]^{\frac{1}{1-\varepsilon}}, 0 \leq \varepsilon \neq 1$$

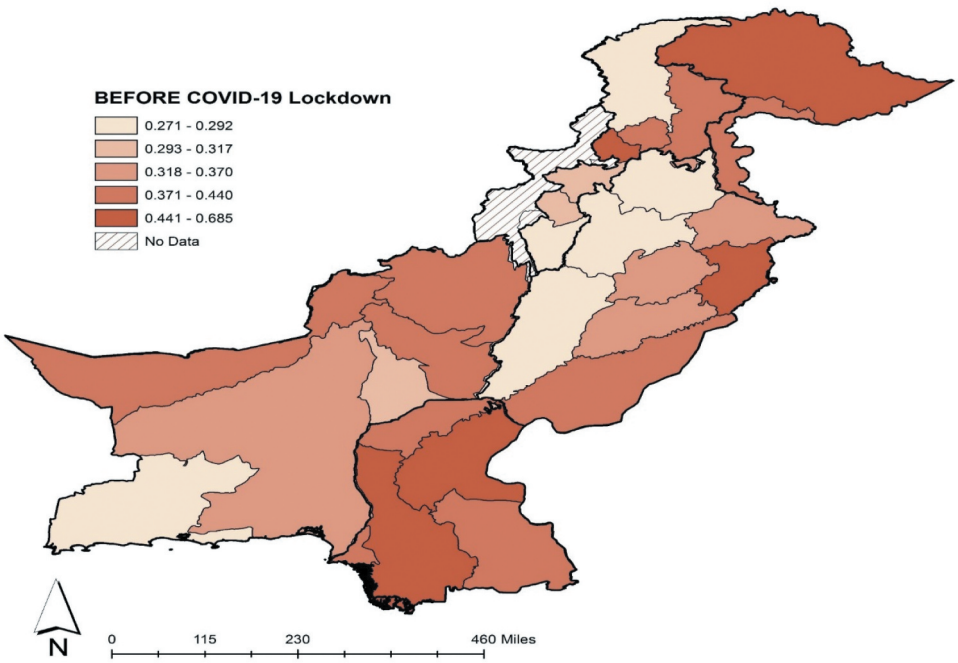

Figure 1. Inequality across Pakistan before COVID-19 lockdown. Note: The map provides information about income inequality (measured by Gini Coefficient) in Pakistan before the COVID-19 lockdown. The data used to construct this map are extracted from the 'Special survey for evaluating socio-economic impact of COVID-19 on wellbeing of people' published by the Pakistan Bureau of Statistics. Each polygon represents a division (administrative-level higher than district). The darker the polygon, the higher the income inequality.

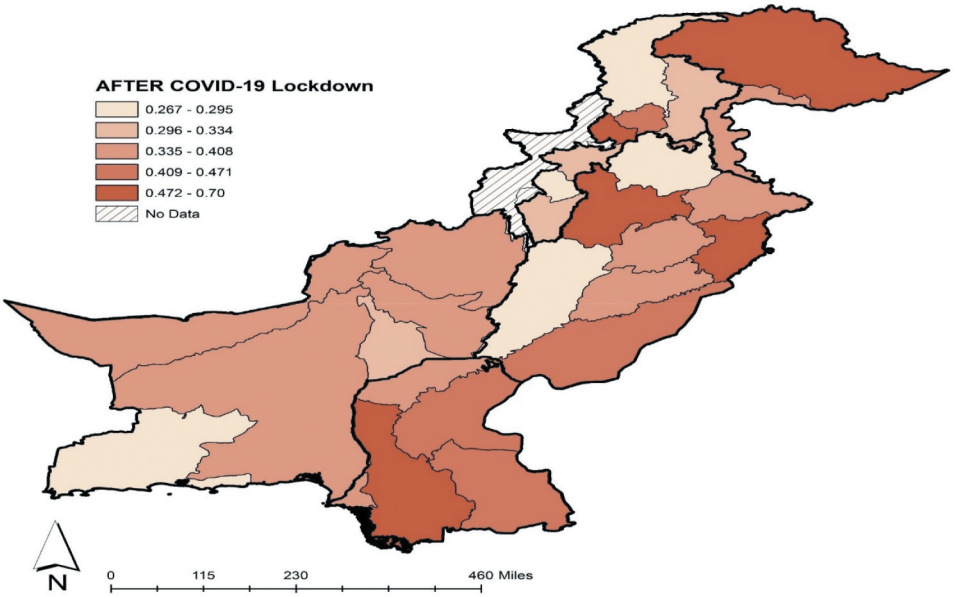

Figure 2. Inequality across Pakistan after COVID-19 Lockdown. Note: The map provides information about income inequality (measured by Gini Coefficient) in Pakistan after the COVID-19 lockdown. The data used to construct this map are extracted from the 'Special survey for evaluating socio-economic impact of COVID-19 on wellbeing of people,' published by the Pakistan Bureau of Statistics. Each polygon represents a division (administrative-level higher than district). The darker the polygon, the higher the income inequality.

$$A_{\varepsilon} = 1 - \frac{1}{\mu}\left(\prod_{i=1}^{N} y_i\right)^{\frac{1}{N}}, \varepsilon = 1$$

While doing our analyses, we use 0.5, 1, and 2 as the respective values of sensitivity parameter (ε). The higher value of the parameter indicates more sensitivity of the Atkinson index to inequalities at the bottom of the income distribution.

Results and discussion

This section presents the findings of the study. Table 1 provides the results of different indices that assess income inequality in Pakistan before and after COVID-19. Column 1 of Table 1 shows statistics before COVID-19 lockdown. It reports that the average monthly income of a working Pakistani before the national lockdown was 22,856 Rupees (137 US dollars) per month, while the median income is 18,000 Rupees (98 US dollars) per month.[3] The results also present interval estimates with a 95% confidence interval. These estimates are obtained by exploiting bootstrapping methods for income inequality measures pioneered by Mills and Zandvakili (1997). When it comes to income inequality, the top 10% of the working population earn 6.79 times higher than the bottom 10% of their counterparts. Similarly, the monthly income of the top 10% of the working population is 2.5 times higher than the population below the median (50%). Similarly, the ratio of earnings of the top 25% of the working class to the bottom 25% is 2.756 which indicates the former group earns 2.756 times higher than the latter. Furthermore, the value of the Gini coefficient is 0.438 indicating a moderate level of

Table 1. Income inequality before and after national lockdown.

	Before COVID-19 (95% CI)	After COVID-19 (95% CI)	Difference (p-value)
Average Monthly Income	22,856.77	22,916.19	−59.42**
	(22,813.17–22,876)	(22,873.84–22,946.2)	(0.0073)
Median Monthly Income	18,000	18,000	0.0073
	(18,000–18,000)	(17,997.04–18,000.96)	(0.3176)
p90/p10	6.793	7.324	−0.5314*
	(6.76–6.82)	(7.30–7.35)	(0.0002)
p90/p50	2.521	2.503	0.0200*
	(2.522–2.525)	(2.517–2.524)	(0.0000)
p75/p25	2.756	2.516	−0.2400*
	(2.751–2.771)	(2.512–2.521)	(0.0000)
Gini Coefficient	0.438	0.453	0.0144*
	(0.437 − 0.439)	(0.452–0.453)	(0.0008)
Atkinson Indices (0.5)	0.171	0.184	0.0137**
	(0.1697–0.1713)	(0.1833 − 0.1851)	(0.0011)
Atkinson Indices (1)	0.294	0.308	0.0140*
	(0.2934–0.2955)	(0.3073–0.3091)	(0.0000)
Atkinson Indices (2)	0.499	0.513	0.0138*
	(0.4982–0.4997)	(0.5119–0.5135)	(0.0000)

Note: This table presents the results of income inequality before and after COVID-19 induced National Lockdown. Columns 1 and 2 of the table provide results of income inequality while Column 3 reports the difference between the first two columns. The analysis presented in this table is based on 8860 observations with 7836 (88.44%) male and 1024 (11.56%) female respondents. For all variables, Bootstrapping methods are employed to compute interval estimates and subsequently perform hypothesis testing for differences before and after COVID-19. Where, * < 0:01; ** < 0:05; and *** < 0:10.
Data Source: Pakistan Bureau of Statistics (2020)

inequality across Pakistan before the COVID-19 lockdown. Lastly, the values of Atkinson Indices are 0.171, 0.294, and 0.499 for the respective values of the inequality aversion parameter.

Column 2 of Table 1 presents results after COVID-19 induced national lockdown in Pakistan. We observe a small increase in an average monthly income after COVID-19 induced lockdown.[4] However, the median income remains the same for the two periods. Moreover, the ratio of the top 10% working population to the bottom 10% population after lockdown is 7.3 indicating that the former group earns 7.3 times more than the latter. The comparison between both periods points to the more skewed income concentration after COVID-19 induced lockdowns. The other reported decile and percentile ratios also indicate an increase in income inequality due to COVID-19 related lockdowns. Similarly, the increase in the value of the Gini coefficient from 0.438 to 0.453 also implies a rise in income inequality. The values of Atkinson Indices also detect an income inequality as a result of pandemic-induced national lockdowns. Furthermore, all these differences are statistically significant.

The increase in inequality can plausibly be attributed to the diversity in the nature of business and economic activities and their heterogeneous exposure to lockdown restrictions. For example, any business related to medicine and food was excluded from these restrictions, while tourism, transportation, education, etc., took the heaviest toll as described earlier. Our finding corroborates with the existing literature on pandemics and economic inequality (Blundell et al., 2020; Elgar et al., 2020; Galletta & Giommoni, 2022; Hanspal et al., 2020; Qian & Fan, 2020; Sayed & Peng, 2021; Witteveen, 2020). However, the results of this study are distant from those of Clark et al., 2021) who find a decline in income inequality in Five European Countries (France, Germany, Italy, Spain, and Sweden) during COVID-19.

Furthermore, Table 2 disaggregates the inequality statistics across provinces. It reports that the ratio of earnings of the top 10% to the bottom 10% has considerably increased in Punjab (indicating an increase in inequality in Punjab), remained the same in KPK, and decreased in Sindh and Baluchistan. Likewise, the Gini coefficient and Atkinson indices also indicate a sharp rise in inequality only for Punjab and Baluchistan provinces. The inequality statistics for KPK and Sindh are roughly similar before and after the national lockdown.

Similarly, Table 3 shows the results for the rural–urban divide. The findings suggest that pandemic-induced lockdown is associated with an increase in inequality across urban areas in Pakistan. However, inequality statistics for rural areas remain the same before and after the national lockdown. These findings complement the fact that the immediate negative impact of COVID-19 was on urban communities and big cities around the world (Du et al., 2020). Additionally, most of the business activity is associated with urban areas, so, intuitively, we observe the negative impact of pandemic-induced lockdown on inequality among the urban population. Nevertheless, most of the rural economic activity is not market-based and dispersed, thus, we observe no significant impact of national lockdown on income disparities across rural populations in Pakistan. Another plausible explanation of heterogeneous effects of lockdown in rural and urban areas can be the difference in compliance and attitude towards COVID-19 lockdown restrictions of people in these areas. Such policies are easier to implement and monitor in urban areas where economic activity is much concentrated and regulated. It was also the priority to focus on such areas.[5]

While it is understandable that not every economic activity was exposed to the lockdown restrictions with the same magnitude as some activities have the potential to be switched to online means. It is reasonable to think that inequality can arise due to differences in potential to be switched online but it's implausible to assume that all the jobs and businesses, which are potentially switchable will indeed be switched. We use the definition provided by Hasan, Rehman, and Zhang (2021) to categorize jobs that can be done remotely from home, thus switchable to online[6], and

Table 2. Income Inequality before and after National Lockdown across Provinces.

	Before COVID-19	After COVID-19	Difference p-value
Punjab			
Average Monthly Income	25,145.15	25,074.64	−70.51** (0.0164)
Median Monthly Income	20,000	19,530.25	−469.75* (0.0002)
p90/p10	6.25	6.714	0.464* (0.0000)
p90/p50	2.5	2.35	−0.15* (0.0000)
p75/p25	2.5	2.5	0*** (0.0633)
Gini Coefficient	0.4295	0.4609	0.03132* (0.0000)
Atkinson Indices (0.5)	0.1637	0.1928	0.0291* (0.0000)
Atkinson Indices (1)	0.2829	0.3202	0.03728* (0.0000)
Atkinson Indices (2)	0.4838	0.5307	0.04694* (0.0000)
KPK			
Average Monthly Income	23,047.91	22,213.7	−834.21** (0.0042)
Median Monthly Income	18,000	18,000	0* (0.0001)
p90/p10	5.031	5.063	0.032* (0.0000)
p90/p50	2.516	2.531	0.015* (0.0000)

(*Continued*)

Table 2. (Continued).

	Before COVID-19	After COVID-19	Difference p-value
p75/p25	2.083	2.273	0.19* (0.0000)
Gini Coefficient	0.4463	0.4414	−0.0049* (0.0000)
Atkinson Indices (0.5)	0.1896	0.1854	−0.0042* (0.0000)
Atkinson Indices (1)	0.2938	0.2900	−0.0038* (0.0000)
Atkinson Indices (2)	0.4399	0.4461	0.00619* (0.0000)
Sindh			
Average Monthly Income	19,896.61	19,912.26	15.65* (0.0000)
Median Monthly Income	15,000	15,012.25	12.25* (0.0000)
p90/p10	8.6	7.5	−1.1* (0.0000)
p90/p50	2.867	2.8	−0.067* (0.0000)
p75/p25	2.778	2.778	0* (0.0000)
Gini Coefficient	0.4335	0.4451	0.0116** (0.0114)
Atkinson Indices (0.5)	0.1552	0.1665	0.01136* (0.0000)
Atkinson Indices (1)	0.2857	0.2964	0.01075* (0.0000)
Atkinson Indices (2)	0.5033	0.5091	0.00581* (0.0000)
Balochistan			
Average Monthly Income	20,748.9	21,586.12	837.22* (0.0000)
Median Monthly Income	16,000	16,008	8* (0.0000)
p90/p10	6	5.875	−0.125* (0.0000)

(*Continued*)

Table 2. (Continued).

	Before COVID-19	After COVID-19	Difference p-value
p90/p50	2.4	2.35	–0.05* (0.0000)
p75/p25	2.917	2.667	–0.25* (0.0000)
Gini Coefficient	0.3701	0.3775	0.00736* (0.0001)
Atkinson Indices (0.5)	0.1115	0.1154	0.00388* (0.0007)
Atkinson Indices (1)	0.2159	0.2227	0.00676* (0.0000)
Atkinson Indices (2)	0.4087	0.4139	0.00514* (0.0007)

Note: Authors' calculations. The analysis presented in this table is based on 8162 observations with 3028 (2649 males and 379 females), 2662 (2386 males and 276 females), 1271 (1131 males and 142 females), 1199 (1111 males and 88 females) respondents in Punjab, Sindh, Khyber-Pakhtunkhwa, and Balochistan, respectively. The remaining proportion of observations belongs to Islamabad, Gilgit-Baltistan, and Azad Kashmir. For all variables, Bootstrapping methods are employed to compute interval estimates and subsequently perform hypothesis testing for differences before and after COVID-19. Where, * < 0:01; ** < 0:05; and *** < 0:10.
Data Source: Pakistan Bureau of Statistics (2020)

Table 3. Income Inequality before and after National Lockdown across Locality.

	Before COVID-19	After COVID-19	Diff (p-value)
Urban Areas			
Average Monthly Income	26,563.42	26,355.67	−207.75* (0.0000)
Median Monthly Income	19,000	18,000	−1000* (0.0000)
p90/p10	6.25	6.714	0.4646* (0.0000)
p90/p50	2.5	2.5	0* (0.0000)
p75/p25	2.462	2.5	0.038* (0.0000)
Gini Coefficient	0.4401	0.4583	0.0182* (0.0000)
Atkinson Indices (0.5)	0.1746	0.1906	0.01594* (0.0000)
Atkinson Indices (1)	0.2966	0.3172	0.02058* (0.0000)
Atkinson Indices (2)	0.5069	0.5298	0.02287* (0.0000)
Rural Areas			
Average Monthly Income	15,191.98	15,347	154.71* (0.0000)
Median Monthly Income	14,978.5	14,942.75	−35.75* (0.0000)
p90/p10	6.867281	6.514868	−0.3524* (0.0000)
p90/p50	2.331358	2.288071	−0.0433* (0.0000)
p75/p25	2.671935	2.515292	−0.1566* (0.0000)
Gini Coefficient	0.3950	0.3980	0.0030* (0.0000)
Atkinson Indices (0.5)	0.1284	0.1358	0.0074* (0.0000)
Atkinson Indices (1)	0.2394	0.2440	0.0046* (0.0000)
Atkinson Indices (2)	0.4261	0.4269	0.0008* (0.0000)

Note: Authors' calculations. The analysis presented in this table is based on 8860 observations with 6089 (5417 male and 672 female), 2771 (2419 male and 352 female) respondents in urban and rural areas, respectively. For all variables, Bootstrapping methods are employed to compute interval estimates and subsequently perform hypothesis testing for differences before and after COVID-19. Where, * < 0:01; ** < 0:05; and *** < 0:10.
Data Source: Pakistan Bureau of Statistics (2020)

estimate the segregated impact of lockdown restrictions on inequality among jobs that are switchable and non-switchable (see Table 4 4 for the detailed results). The estimates suggest that inequality has been considerably increased in the jobs that can be done remotely from home. On the other hand, there is an insignificant increase in inequality for jobs that cannot be done remotely from home. This result makes sense as most Pakistanis do not have access to an internet connection to perform the tasks remotely Hasan et al. (2021). This inequality in opportunities to go online contributes to an increase in income disparities across occupations that require internet access to get things done from home.

Table 4. Income Inequality before and after National Lockdown across Job Nature.

	Before COVID-19	After COVID-19	Diff (p-value)
Switchable			
Average Monthly Income	31,561.91	32,061.62	499.71* (0.0000)
Median Monthly Income	24,679.75	24,167.1	−512.65* (0.0000)
p90/p10	8.125	9.286	1.161* (0.0000)
p90/p50	2.682	2.764	0.0824* (0.0000)
p75/p25	3.077	3.333	0.256* (0.0000)
Gini Coefficient	0.47988	0.50209	0.02221* (0.0000)
Atkinson Indices (0.5)	0.19789	0.22048	0.02259* (0.0000)
Atkinson Indices (1)	0.34273	0.37144	0.02871* (0.0000)
Atkinson Indices (2)	0.5734	0.60822	0.03482* (0.0000)
Not Switchable			
Average Monthly Income	20,255.87	20,170.31	−85.56* (0.0004)
Median Monthly Income	17,952.25	16,847.77	−1104.48* (0.0000)
p90/p10	6.5738	6.6480	0.0742* (0.0000)
p90/p50	2.3816	2.2275	−0.15411* (0.0000)
p75/p25	2.3428	2.4939	0.15105* (0.0000)
Gini Coefficient	0.40583	0.41415	0.00832* (0.0000)
Atkinson Indices (0.5)	0.14911	0.15698	0.00787* (0.0000)
Atkinson Indices (1)	0.26034	0.26852	0.00818* (0.0000)
Atkinson Indices (2)	0.45803	0.46451	0.00648* (0.0000)

Note: Authors' calculations. The analysis presented in this table is based on 8860 observations with 7051 (6313 male and 738 female), 1809 (1523 male and 286 female) respondents in switchable and unswitchable areas, respectively. For all variables, Bootstrapping methods are employed to compute interval estimates and subsequently perform hypothesis testing for differences before and after COVID-19. Where, * < 0:01; ** < 0:05; and *** < 0:10.
Data Source: Pakistan Bureau of Statistics (2020)

Finally, we report the income inequality across different occupational groups classified by the Pakistan Standard Classification of Occupations (PSCO) in Table 5 5. Out of 45 reported occupation groups, we notice an increase in income inequality in 27 of them. Furthermore, Legal, social and cultural professionals, Business and administration professionals, Protective services workers, Refuse workers and other elementary workers, and Handicraft and printing workers are the top five occupational groups where we observe the highest increase in income inequality associated with the pandemic-induced lockdown. Contrarily, occupational groups such as Commissioned armed forces officers, Information and communications technicians, Administrative and commercial managers, General and keyboard clerks, and Subsistence farmers, fishers, hunters, and gatherers are observed to have achieved equity within their respective occupational groups due to national lockdown.

Table 5. Income inequality before and after national lockdown across occupations.

Code	Occupation	Before COVID-19	After COVID-19	Difference (p-value)
1	Commissioned armed forces officers	0.372	0.190	−0.182* (0.0000)
2	Non-commissioned armed forces officers	0.219	0.225	0.005 (0.2372)
3	Armed forces occupations, other ranks	0.198	0.187	−0.011 (0.2701)
11	Chief executives, senior officials and legislators	0.337	0.343	0.007* (0.0000)
12	Administrative and commercial managers	0.413	0.342	−0.070* (0.0000)
13	Production and specialised services managers	0.474	0.484	0.010* (0.0006)
14	Hospitality, retail, and other services managers	0.638	0.646	0.008* (0.0023)
21	Science and engineering professionals	0.254	0.272	0.018* (0.0000)
22	Health professionals	0.623	0.625	0.001* (0.0375)
23	Teaching professionals	0.445	0.443	−0.002 (0.2922)
24	Business and administration professionals	0.463	0.548	0.086* (0.0000)
25	Software and applications developers and analysts	0.340	0.341	0.002* (0.0852)
26	Legal, social and cultural professionals	0.461	0.559	0.098* (0.0000)
31	Science and engineering associate professionals	0.288	0.301	0.014* (0.0000)
32	Health associate professionals	0.340	0.349	0.009* (0.0000)
33	Business and administration associate professionals	0.351	0.377	0.025* (0.0000)
34	Legal, social, cultural and related associate professionals	0.207	0.237	0.030 (0.8947)
35	Information and communications technicians	0.306	0.227	−0.079* (0.0000)
41	General and keyboard clerks	0.236	0.184	−0.052* (0.0000)
42	Customer services clerks	0.231	0.246	0.014* (0.0000)
43	Numerical and material recording clerks	0.286	0.288	0.002* (0.0000)
44	Other clerical support workers	0.245	0.255	0.010 (0.0179)
52	Sales workers	0.343	0.358	0.015* (0.0000)
51	Personal Services	0.377	0.393	0.016* (0.0000)
53	Personal care workers	0.438	0.435	−0.003*** (0.0987)
54	Protective services workers	0.269	0.351	0.082* (0.0000)
61	Market-oriented skilled agricultural workers	0.541	0.560	0.019* (0.0000)
62	Market-oriented skilled forestry, fishery, and hunting workers	0.252	0.234	−0.018* (0.0000)
63	Subsistence farmers, fishers, hunters, and gatherers	0.492	0.462	−0.030* (0.0000)
71	Building and related trades workers, excluding electricians	0.318	0.325	0.007 (0.3704)
72	Metal, machinery and related trades workers	0.398	0.418	0.020* (0.0000)

(*Continued*)

Table 5. (Continued).

Code	Occupation	Before COVID-19	After COVID-19	Difference (p-value)
73	Handicraft and printing workers	0.353	0.384	0.032* (0.0000)
74	Electrical and electronic trades workers	0.347	0.330	−0.017* (0.0000)
75	Food processing, woodworking, garment, and other craft and related	0.383	0.411	0.028* (0.0000)
81	Stationary plant and machine operators	0.328	0.346	0.018* (0.0000)
82	Assemblers	0.283	0.279	−0.004 (0.1025)
83	Drivers and mobile plant operators	0.264	0.264	−0.000 (0.1758)
91	Cleaners and helpers	0.343	0.325	−0.018* (0.0000)
92	Agricultural, forestry and fishery labourers	0.379	0.363	−0.017* (0.0000)
93	Labourers in mining, construction, manufacturing and transport	0.288	0.271	−0.017* (0.0000)
94	Food preparation assistants	0.205	0.224	0.018* (0.0001)
95	Street and related sales and service workers	0.273	0.268	−0.005* (0.0000)
96	Refuse workers and other elementary workers	0.239	0.299	0.060* (0.0000)

Note: Authors' calculations.
For all variables, Bootstrapping methods are employed to compute interval estimates and subsequently perform hypothesis testing for differences before and after COVID-19. Where, * <0:01; ** <0:05; and *** <0:10.
Data Source: Pakistan Bureau of Statistics (2020)

Conclusions

This paper contributes to the existing literature on the COVID-19 pandemic by estimating the association of pandemic-induced national lockdown and income inequality in Pakistan. To do so, we utilized a nationally representative survey, 'Special Survey for Evaluating Socio-Economic Impact of COVID-19 on Wellbeing of People,' conducted by the Pakistan Bureau of Statistics (2020). The findings of the study indicate an increase in income inequality due to COVID-19 induced economic shutdowns in Pakistan. The disaggregated analysis suggests an increase in inequality only in Punjab and Balochistan but not in KPK and Sindh. Likewise, our study observes an increase ncomee equality in urban areas given the high exposure of urban areas (areas with a higher concentration of business-related activities). Moreover, the findings also suggest that COVID-19 induced lockdown affected both income inequality in switchable and unswitchable, while it is observed that inequality in unswitchable jobs among the 90th percentile and 50th percentile has decreased after COVID-19 induced lockdown. Lastly, it is observed that lockdowns have a heterogeneous effect on different occupational groups.

This study has important policy implications for Pakistan and other countries where large-scale restrictions are imposed in response to the COVID-19 pandemic. It is observed that such policies achieve the targeted objectives, however, they may produce externalities in the form of income inequality. An institution can be set up, which can work as a subdivision of the Planning Commission, to administer and form policies to mitigate these adverse effects of lockdowns. Such an institute should work in close collaboration with (or can be made an integral part of) the institution which administers the spread of COVID-19 and synergizes the national efforts to curb it such as the National Command and Operation Center (NCOC) in Pakistan. As the pandemic is not over yet, policymakers should therefore use a combination of policies to curb not only COVID-19 spread but also to protect the livelihood of the vulnerable population.

Notes

1 https://www.dawn.com/news/1543683/
2 The original phrasing of the question is, 'Did (respondent) do or help in any work for pay, profit or family gains or have a job or enterprise such as a shop, business, farm, or service establishment (fixed or mobile) During Jan, to March 2020? If yes then, Average Monthly income during that time?' The same question was asked during the lockdown income and after lockdown income.
3 Throughout our analysis, we used average exchange rate in April 2019 (167 PKR = 1 US Dollar).
4 Readers should be aware of the fact that this paper does not 'claim' the causal impact of national lockdown on income inequality. Rather, we are providing suggestive evidence.
5 Our gender-specific results suggest that the average monthly income of men is 2 times the average monthly income of their women counterparts. Similarly, the median income of men is 3 times that of the women. Nevertheless, these statistics remained the same before and after the national lockdown. However, women – who already earn less and have more unequal incomes compared to men – are observed to have more skewness as a result of national lockdown. Percentile ratios indicate that the top 10% of working females earn 13 times higher than the females from the bottom 10% of the population distribution before lockdown. But the same ratio rose to 16.2 after the lockdown. This finding indicates that the most vulnerable among the vulnerable segments of society, such as females, are more adversely affected by the lockdown restrictions. These findings are in-line with that of Blaskó, Papadimitriou, and Manca (2020) who noted, 'women are risking to pay a higher price for the crisis than men' (Blaskó et al., 2020, p. 16). This result also highlights the need to pay deliberation to the vulnerable segments of society while designing and implementing public policy in response to a crisis such as the COVID-19 pandemic. Readers who are interested to learn more about gender-specific analysis can request the detailed results by writing to the corresponding author.
6 The job is regarded as switchable.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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